

Supplementary Material for “Learning-Aware Lyapunov Resource Control for Correlated Multi-Agent Markov Learning”

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I. RESOURCE-GEOMETRY PROOFS

A. Proof of Proposition 1

Under a message-only budget, the number of independent rounds is $T = B_{\text{msg}}/(h+q)$ up to integer rounding. Substituting this horizon into (11) of the main paper and dropping the common positive factor $(\sigma_c^2 + \sigma_e^2)/B_{\text{msg}}$ gives (13). Expanding it yields

$$F(q; \rho, h) = h\rho + 1 - \rho + q\rho + \frac{h(1-\rho)}{q}. \quad (1)$$

For $q > 0$,

$$F'(q) = \rho - \frac{h(1-\rho)}{q^2}, \quad F''(q) = \frac{2h(1-\rho)}{q^3} \geq 0. \quad (2)$$

The unique interior stationary point is therefore (14), and convexity makes it the continuous minimum. Clipping handles the interval boundary. For integer q , discrete convexity implies that a minimizer is a feasible neighbor of the clipped continuous optimum.

B. Geometric temporal-correlation sum

The identity

$$\sum_{s=1}^{N-1} (N-s)r^s = N \sum_{s=1}^{N-1} r^s - \sum_{s=1}^{N-1} sr^s \quad (3)$$

together with the finite geometric-series derivative gives (15) of the main paper. This term is nondecreasing in $r \in [0, 1]$. Because $r = \lambda^b$ is nonincreasing in b , physical spacing decreases correlation cost at a fixed N . Equation (8) of the main paper shows why the same monotonicity need not hold after charging the environment clock.

C. Exact risk of the running-mean learner

Let $\bar{X}_k = q^{-1} \sum_{i=1}^q X_{i,k}$ under the common-factor model. After subtracting its target mean, the stationary covariance is

$$\text{Cov}(\bar{X}_k, \bar{X}_\ell) = \theta_j r^{|k-\ell|} + \frac{1}{q} \mathbf{1}\{k = \ell\}. \quad (4)$$

The registered unit pseudo-observation gives the estimator

$$\hat{\vartheta}_N = \frac{\hat{\vartheta}_0 + \sum_{k=1}^N \bar{X}_k}{N+1}, \quad (5)$$

with unit initial squared error independent of the observations. Summing every diagonal and off-diagonal covariance yields

$$\mathbb{E}(\hat{\vartheta}_N - \vartheta)^2 = \frac{1}{(N+1)^2} \left[1 + \frac{N}{q} + \theta_j \left\{ N + 2 \sum_{s=1}^{N-1} (N-s)r^s \right\} \right], \quad (6)$$

which proves (10) of the main paper.

D. Proof of Proposition 3

Apply an orthogonal rotation across actors at every retained time. The $q-1$ spatial contrasts contain only unit-variance private noise and have the same law under both hypotheses. The normalized spatial mean $Y_k = q^{-1/2} \mathbf{1}^\top X_k$ has covariance

$$\Sigma_j = I_n + q\theta_j T_n(r). \quad (7)$$

Thus the contrasts cancel from every likelihood ratio, and the divergence is that of two centered Gaussian laws with covariances Σ_0 and Σ_1 . The matrices are polynomials in the same symmetric Toeplitz matrix and hence simultaneously diagonalizable. The Gaussian covariance-divergence formula

$$\text{KL}(P_0 \| P_1) = \frac{1}{2} \{ \text{tr}(\Sigma_1^{-1} \Sigma_0) - n - \log \det(\Sigma_1^{-1} \Sigma_0) \} \quad (8)$$

therefore yields (31) and (32) of the main paper after substituting the shared eigenvectors and eigenvalues $1 + q\theta_j \zeta_\ell$. The centered-Gaussian Bhattacharyya formula gives (33) under the same diagonalization. Its likelihood-ratio bound makes each directional error at most $\exp(-I_B)$, justifying the probe-length rule in Section VI-B.

II. EXPERIMENTAL ESTIMANDS

At registered charged-resource fractions $0 = u_0 < \dots < u_L = 1$, let $E_{s,p}^x(u_\ell)$ be the seed- s , policy- p squared parameter or initial-state return-estimation error, normalized by its value at u_0 . The trajectory area under the curve is

$$\text{AUC}_{s,p}^x = \sum_{\ell=1}^L \frac{u_\ell - u_{\ell-1}}{2} [E_{s,p}^x(u_\ell) + E_{s,p}^x(u_{\ell-1})], \quad (9)$$

where $x \in \{\text{par}, \text{ret}\}$. The resource axis charges skipped transitions, actor messages, and terminal delay.

Intervals are the 2.5–97.5 percentiles of 2,000 frozen-seed bootstrap replicates that resample paired seeds and recompute the geometric-mean risk ratio over all registered cells; pairing preserves common-random-number variance reduction without treating checkpoints as independent. Finite-value and cumulative-budget checks precede aggregation, and the fixed- q baseline is selected once on the disjoint development population and frozen for confirmation.